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ABSTRACT

Organic polymers in the polymer matrix are normally insulators. There is a huge demand for the production of conductive polymers, especially from the electronic industry. The insertion of highly conductive inorganic fillers into the polymer matrix is the most common way to improve the conductivity of a polymer. But to date, to achieve high conductivity, high loading of the fillers is needed, sacrificing the lightweight, excellent processability and low cost of polymeric materials. The goal of this paper is to evaluate the effect of filler on various parameters such as dielectric constant, dielectric loss and conductive property of Polyvinyl alcohol (PVA)/Ferrous Ferric Oxide (Fe₃O₄) for a frequency range varying between 20 Hz and 10 MHz. PVA/Fe₃O₄ nanocomposite was fabricated by using solvent casting method. Variation in properties was measured by impedance analyser. This paper concludes that on adding 0.8 gm of Fe₃O₄ to PVA, the best results of conductivity, dielectric constant and dissipation factor. Relevant parameter like conductivity of the membranes was highly affected by Fe₃O₄ content and the conductivity increases with increase in Fe₃O₄ content. The addition of Fe₃O₄ as a filler to polymer matrix enhances the mentioned parameters, mechanical strength and magnetic behaviour. The ohmic behaviour was analyzed by Cole-cole plot. The nanocomposite formed by PVA/Fe₃O₄ because of its parameters can be used in biomedical applications, drug delivery and biosensors.

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1. Introduction

Polymers are known for their insulating properties, which can be adjusted by compounding with electrically conductive filler to become relatively good electrical conductors. The application of conducting polymer composites are temperatures sensors, self-regulating heaters, over current, current clipper, over temperature protection devices, biomedical applications and antistatic materials for electromagnetic interference (EMI) shielding of electronic devices [1,2]. PVA is a water-soluble, non-toxic synthetic polymer, which due to its excellent film forming, emulsifying, and adhesion properties has been used in pharmaceutical and biomedical application [3,4]. PVA was prepared from polyvinyl-ester and has been commonly used in biomedical and industrial applications. This polymer was recognized recently as biodegradable despite its synthetic character [5]. It is pliant but flexible, strong and functions as high oxygen and aroma barrier. Fillers like ferrites and metal oxides

are used highly important to provide improved magnetic and electrical properties [6]. Fe₃O₄ is also known as Magnetite and is black in colour. It is used in the Haber process as a catalyst. Magnetite (Fe₃O₄) is preferred and most suitable to be used with polymers due to its typical magnetic and electrical properties [7]. Attractive iron oxide (Fe₃O₄) are likewise utilized as a medication conveyance framework in light of its biodegradability and biocompatibility to living cells [8]. In previous studies, it has been found that the iron oxide containing nanocomposite show superparamagnetic behaviour used for bio-medical application [9]. Nanocomposite of Fe₃O₄ and PVA were also synthesized by coprecipitation method for drug delivery system [10]. Flexible polymer membrane used in electronics devices were also fabricated by mixing PVA-GL (glycerol) combination with Fe₃O₄ [11]. PVA/Fe₃O₄ ferrogels were also fabricated by PVA/Fe₃O₄ by freezing-thawing (F-T) cycle technology used for bio-medical application [12,13]. PVA/Fe₃O₄ blend membrane are also used as novel glucose biosensor. The matrix of PVA when dispersed with Fe₃O₄ between enzyme and electrode [14]. The proposed paper deals with the concept of fillers being added with polymer to enhance its

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conductivity, dielectric and reduced dissipation factor. This nanocomposite formed after addition of filler increased its mechanical strength [9].

2. Experimental setup

2.1. Materials

Material used for membrane fabrication is Polyvinyl alcohol (PVA) and Ferrous Ferric Oxide (Fe_3O_4). PVA is a white odourless man-made polymer which is hydrophilic nature. Ferrous Ferric Oxide (Fe_3O_4) is a black oxide powder which shows magnetic properties and behaves as filler. Distilled water is used as solvent. These materials were purchased from Aldrich.

2.2. Fabrication technique

Membrane preparation procedure is shown in Fig. 1. The schematic diagram of PVA and Fe_3O_4 is shown in Fig. 1(a) and (b). Fig. 1 (c) and (d) indicate the manufacturing phase of a membrane. In Fig. 1(c), the various combinations of PVA and Fe_3O_4 are poured into a conical beaker and dry stirred until fully combined and then added with distilled water to make a solution. Fig. 1(c) shows the evaporation of solvent. The mixing and continuous heating process is carried out for around 100°C for 3 h. After these two methods, Distilled water is added in the beaker. It is again heated 3 h at 100°C until PVA is dissolved and mixed uniformly with Fe_3O_4 . This contributes to creation of a pure black liquid. After performing this process, the solution is poured into the petri dish and dried at 100°C for 3 h. Fig. 1(d) demonstrates the creation of membrane involving drying and removal of the membrane from the petri dish that provides final black membrane of PVA/ Fe_3O_4 . Fig. 1(e) shows the final black membrane. Same method is utilized to make different membrane with different suitable ratio of PVA and Fe_3O_4 . Different ratios of PVA/ Fe_3O_4 are represented in Table 1.

3. Results and discussion

Impedance Analyzer E4900A (Key sight technologies) is used to calculate different impedance properties at ambient temperature. Impedance Analyzer evaluates the relevant parameters.

3.1. Dielectric constant

The equation (1) shows the dielectric constant formula which is used

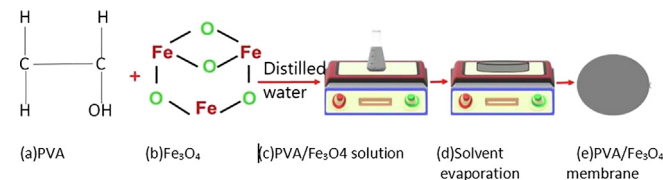


Fig. 1. (a) Schematic diagram of PVA and Fe_3O_4 (b)(c)(d) Procedure for Membrane Fabrication.

Table 1
Different ratio PVA and Fe_3O_4 for membrane fabrication.

Ratio	Polyvinyl alcohol (PVA)	Fe_3O_4 (gm)	DI water (ml)
50/50	1	1	19
60/40	1.2	0.8	19
80/20	1.6	0.4	19
90/10	1.8	0.2	19

$$\varepsilon = \frac{l \times C}{A \times \varepsilon_0} \quad (1)$$

Fig. 2 shows the frequency related change in dielectric constant. In Fig. 2, the value of ε decreases smoothly with increasing frequency for all the samples. The dielectric constant has an elevated value at low frequencies. Electric field shift is actually followed by dipoles at low frequencies [15,16]. The dielectric value decreases at high frequencies, because there is no excess dipole molecules and there is no sufficient time for ion diffusion to adjust itself in direction of electric field. High dielectric constant in PVA/ Fe_3O_4 (60/40) sample makes it useful for use in actuators, resonators, sensors and battery separators.

3.2. Conductivity

The conductivity was calculated by using equation (2).

$$\sigma = \omega \times \varepsilon' \times \varepsilon_0 \times \tan \delta \quad (2)$$

Fig. 3 indicates conductivity difference with frequency. The conductivity enhances with the increase in frequency. Conductivity enhances with enhancement in addition of Fe_3O_4 content. This occurs due to increase in carrier density or because of increase in

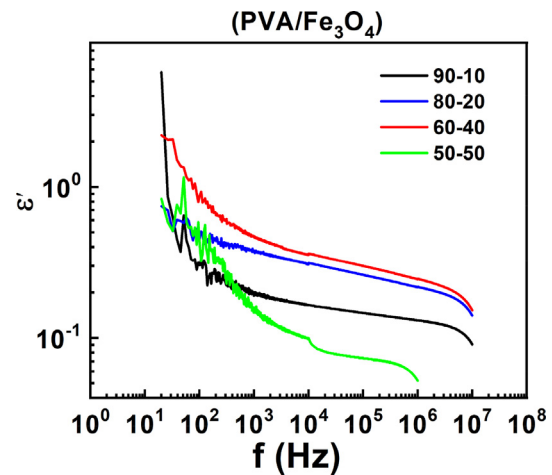


Fig. 2. Graph representing Dielectric with respect to frequency of different membrane.

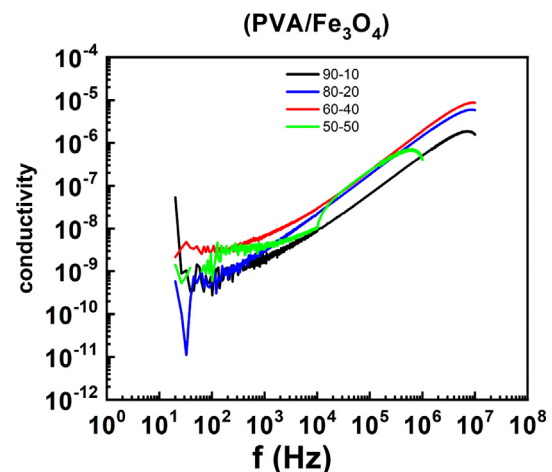


Fig. 3. Graph representing Conductivity with respect to frequency range of different membrane.

mobility of electrons in membrane [17]. This improves the concentration of the nanoparticles which increases the conductivity and also enhances the charge movement. The charge transport is sluggish in the low frequency, indicating the low conductivity and as the frequency increases the charge movement increases as shown in Fig. 3.

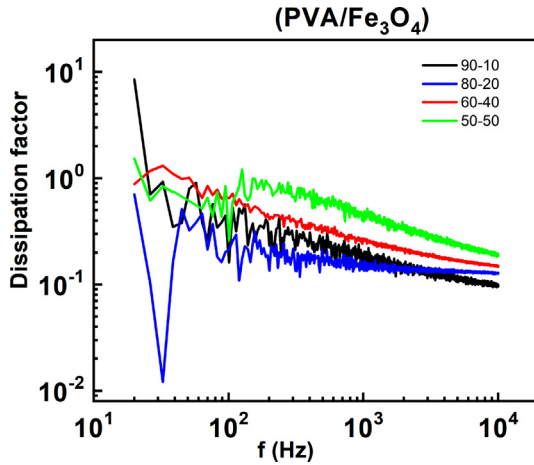


Fig. 4. Graph representing Dissipation factor with respect to frequency range of different membrane.

3.3. Dissipation factor

Fig. 4 indicates dissipation factor (dielectric loss) of different composition of PVA/Fe₃O₄ membrane with frequency. Fig. 4 shows that dissipation factor decreases when there is a frequency rise and this drop in dissipation increases on further increase in frequency. This happens because as electric oscillations are decreased when charge induced fails to move towards the reversing field as the frequency increases [18]. Under the influence of an applied electric field, the charge conducting carriers existing in the samples can travel upto specific distance. The electrodes avoid these type of carriers and this leads to an environment where dissipation factor would be increasing at lower frequencies.

3.4. Impedance analysis

Different ratio of PVA/Fe₃O₄ membrane's cole-cole plots are shown in Fig. 5. The imaginary part of the impedance Z'' indicates energy stored in the above samples. Complex impedance analysis offers details about relaxation frequency, and capacitance of grain boundary, conductivity and grain boundary resistance. To study the sample's electrical behaviour, Cole-Cole plot is used [19]. In cole-cole plot, the left intercept of the curve cutting the X-axis is known as the real ohmic resistance of the membrane and the curve on right side tell resistance of charge transfer.

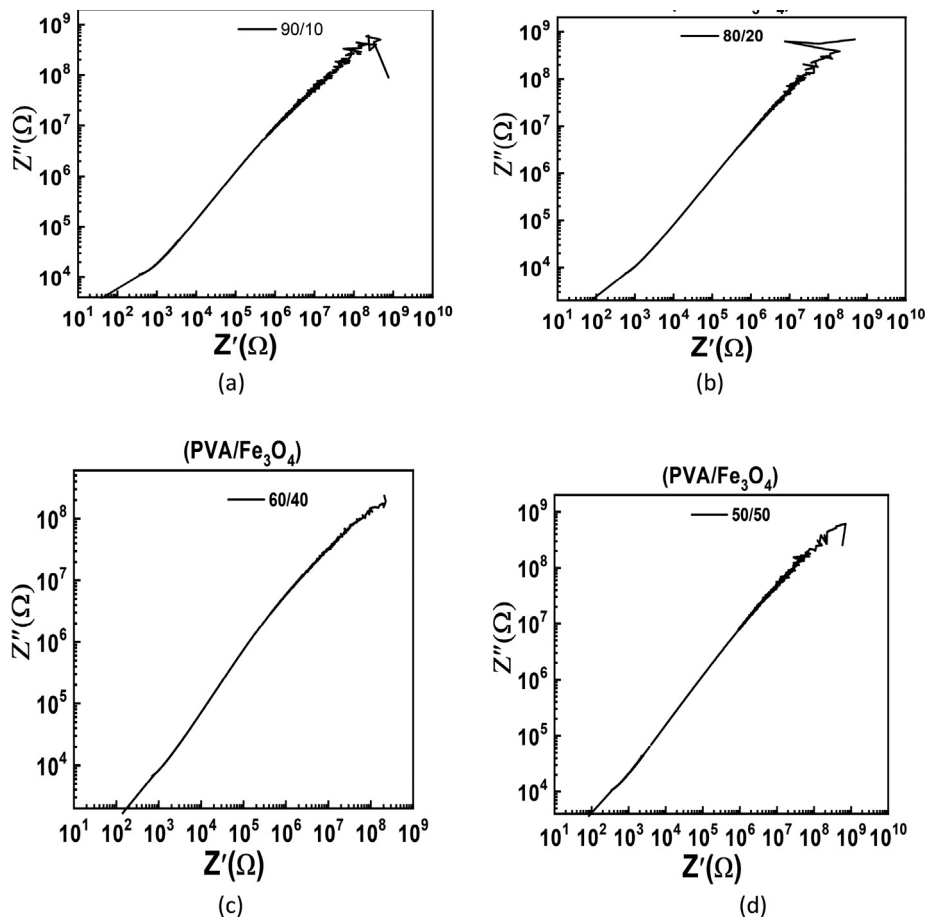


Fig. 5. Graph representing Cole-cole plot of (a) 90/10 (b) 80/20 (c) 60/40 (d) 50/50 ratio PVA/Fe₃O₄ membrane.

4. Conclusion

The nanocomposite membrane composed of PVA/Fe₃O₄ was developed by solvent casting method. This nanocomposite membrane visualized good dielectric with the addition of black oxide filler. The addition of Fe₃O₄ metal oxide filler increased the dielectric constant, increased conductivity, reduced the dissipation factor. This kind of black membrane is very much helpful for TFT, battery, electro-chemical capacitor and wearable sensing devices [20,21]. It is clear from the graph that conductivity and dielectric increase with increasing composition of Fe₃O₄ percent. PVA/Fe₃O₄ (60/40) is the best composition showing highest dielectric constant, good conductivity and low dielectric loss.

CRedit authorship contribution statement

Sribidhya Mohanty: Conceptualization, Data curation, Formal analysis, Investigation. **Varij Panwar:** Software, Supervision, Validation, Visualization, Funding acquisition. **Priya Khanduri:** Methodology, Project administration, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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